

东北大学资源与土木工程学院

作业报告

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1.12 VISUAL IMAGE INTERPRETATION

1.12 视觉图像解译

The hue cancellation method is based on the observation that when certain colors are mixed together, the resulting colors are not what would be intuitively expected. For example, when red and green are mixed together, they produce yellow, not reddish green. (For further information, see Robinson et al., 1995.)

色相抵消法的依据是，在观察某些颜色混合时，产生的颜色并不是我们直观预料到的。例如，红色和绿色混合会产生黄色，而不是红绿色。（更多信息请参考 Robinson 等人的研究，1995 年）

In the remainder of this discussion, we focus on the trichromatic theory of color vision. Again, this theory is based on the concept that we perceive all colors by synthesizing various amounts of just three (blue, green, and red).

在后续讨论中，我们的重点是视觉三原色理论。该理论的核心观点是人对所有颜色的感知都来自于对三原色（蓝、绿、红）的不同比例合成。

Blue, green, and red are termed *additive primaries*. Plate 3a shows the effect of projecting blue, green, and red light in partial superimposition. Where all three beams overlap, the visual effect is white because all three of the eyes' receptor systems are stimulated equally. Hence, white light can be thought of as the mixture of blue, green, and red light. Various combinations of the three additive primaries can be used to produce other colors. As illustrated, when red light and green light are mixed, yellow light is produced. Mixture of blue and red light results in the production of magenta light (bluish-red). Mixing blue and green results in cyan light (bluish-green).

蓝、绿、红被称为**加色基色**。图 3a 展示了蓝、绿、红光部分重叠投影的效果：三束光完全叠加的区域呈现白色，这是因为人眼的三种感光系统受到了相等的刺激。因此，白光可被理解为蓝、绿、红三种色光的混合色。通过调整三种加色基色的组合比例能产生其他各种颜色：红光与绿光混合生成黄光，蓝光与红光混合生成品红光（蓝红色），蓝光与绿光混合则生成青色光（蓝绿色）。

Yellow, magenta, and cyan are known as the *complementary colors*, or *complements*, of blue, green, and red light. Note that the complementary color for any given primary color results from mixing the remaining two primaries.

黄、品红、青被称为蓝、绿、红光的**补色**。需要注意的是，任意一种加色基色的补色，都是由另外两种加色基色混合而成的。

Like the eye, color television and computer monitors operate on the principle of additive color mixing through use of blue, green, and red elements on the screen. When viewed at a distance, the light from the closely spaced screen elements forms a continuous color image.

彩色电视和电脑显示器与人的视觉原理一样，也采用加色混合原理——屏幕通过蓝、绿、红三色像素发光，当从一定距离观看时，密集排列的像素发出的光会融合形成连续的彩色图像。

Whereas color television and computer monitors simulate different colors through *additive* mixture of blue, green, and red *lights*, color film photography is based on the principle of *subtractive* color mixture using superimposed yellow, magenta, and cyan *dyes*. These three dye colors are termed the *subtractive primaries*, and each results from subtracting

one of the additive primaries from white light. That is, yellow dye absorbs the blue component of white light. Magenta dye absorbs the green component of white light. Cyan dye absorbs the red component of white light.

彩色电视和显示器通过蓝、绿、红三色光的**加色**混合模拟各种色光，而彩色胶片摄影则是基于**减色**混合原理。它利用黄、品红、青三色染料的叠加来显示颜色。这三种染料被称为**减色基色**，每种染料从白光中吸收一种加色基色：黄色染料吸收白光中的蓝光成分，品红染料吸收绿光成分，青色染料吸收红光成分。

The subtractive color-mixing process is illustrated in Plate 3*b*. This plate shows three circular filters being held in front of a source of white light. The filters contain yellow, magenta, and cyan dye. The yellow dye absorbs blue light from the white background and transmits green and red. The magenta dye absorbs green light and transmits blue and red. The cyan dye absorbs red light and transmits blue and green. The superimposition of magenta and cyan dyes results in the passage of only blue light from the background. This comes about because the magenta dye absorbs the green component of the white background, and the cyan dye absorbs the red component. Superimposition of the yellow and cyan dyes results in the perception of green. Likewise, superimposition of yellow and magenta dyes results in the perception of red. Where all three dyes overlap, all light from the white background is absorbed and black results.

图 3*b* 展示了减色混合的过程：三个黄、品红、青色的圆形滤光片置于白光光源前。黄色滤光片吸收白光中的蓝光，透过绿光和红光；品红滤光片吸收绿光，透过蓝光和红光；青色滤光片吸收红光，透过蓝光和绿光。品红与青色滤光片叠加时，只有蓝光能透过（品红吸收绿光，青色吸收红光）；黄色与青色滤光片叠加时，人眼看到为绿色；黄色与品红滤光片叠加时呈现红色；而三个滤光片完全叠加时，白光中的所有色光均被吸收，最终呈现黑色。

In color film photography, and in color printing, various proportions of yellow, magenta, and cyan dye are superimposed to control the proportionate amount of blue, green, and red light that reaches the eye. Hence, the subtractive mixture of yellow, magenta, and cyan dyes on a photograph is used to control the additive mixture of blue, green, and red light reaching the eye of the observer. To accomplish this, color film is manufactured with three emulsion layers that are sensitive to blue, green, and red light but contain yellow, magenta, and cyan dye after processing (see Section 2.4).

在彩色胶片摄影和彩色印刷中，人们会叠加不同比例的黄、品红、青染料来控制人眼看到的蓝、绿、红三色光的比例。因此，照片上黄、品红、青染料的减色混合，本质是为了调控人们眼中蓝、绿、红三色光的加色混合效果。为了实现这一目的，彩色胶片制作时采用三层乳剂涂层——这些涂层分别对蓝、绿、红光敏感，经冲洗处理后会生成黄、品红、青三种染料（详见 2.4 节）。

In digital color photography, the photosites in the detector array are typically covered with a blue, green, or red filter, resulting in independent recording of the three additive primary colors (see Section 2.5).

在数字彩色摄影中，探测器阵列的感光单元通常覆盖着蓝、绿或红色滤光片，以此来实现对三种加色基色的独立感光（详见 2.5 节）。

When interpreting color images, the analyst should keep in mind the relationship between the color of a feature in the imagery, the color mixing process that would produce that color, and the sensor's wavelength ranges that are assigned to the three primary colors used in that mixing process. It is then possible to work backwards to infer the spectral

properties of the feature on the landscape. For example, if a feature in a false-color image has a yellow hue when displayed on a computer monitor, that feature can be assumed to have a relatively high reflectance in the wavelengths that are being displayed in the monitor's red and green color planes, and a relatively low reflectance in the wavelength displayed in blue (because yellow results from the additive combination of red plus green light). Knowing the spectral sensitivity of each of the sensor's spectral bands, the analyst can interpret the color as a spectral response pattern (see Section 1.4) that is characterized by high reflectance at two wavelength ranges and low reflectance at a third. The analyst can then use this information to draw inferences about the nature and condition of the feature shown in the false-color image.

在解读彩色图像时，分析人员需牢记以下三点：图像中地物的颜色、生成该颜色的混色原理、以及传感器波长范围与混色所用的三原色的对应关系。在此基础上，分析人员可以通过反向推导得出地物的光谱特性。例如，若假彩色图像中的某一地物在电脑显示器上呈现黄色，说明该物体在显示器红、绿通道所对应的波长范围内反射率较高，而在蓝通道对应的波长范围内反射率较低（因为黄色由红光与绿光加色混合而成）。结合传感器各光谱波段的光谱灵敏度，分析人员可将该颜色解读为一种光谱的响应模式（详见 1.4 节）——其特征是两个波长范围的反射率较高，而第三个波长范围的反射率较低。利用这一信息，分析人员可以进一步推断假彩色图像中地物的性质与状态。

Temporal Aspects of Image Interpretation

图像解译的时间维度

The temporal aspects of natural phenomena are important for image interpretation because such factors as vegetative growth and soil moisture vary during the year. For crop identification, more positive results can be achieved by obtaining images at several times during the annual growing cycle. Observations of local vegetation emergence and recession can aid in the timing of image acquisition for natural vegetation mapping. In addition to seasonal variations, weather can cause significant short-term changes. Because soil moisture conditions may change dramatically during the day or two immediately following a rainstorm, the timing of image acquisition for soil studies is very critical.

因为植被生长、土壤湿度等因素会随季节变化，所以自然现象的时间维度对图像解译非常重要。在进行作物识别时，在每年生长周期内多次获取影像，能够得到更准确的结果。通过观察当地植被的萌发与死亡的规律，可以辅助确定植被制图的影像获取时间。除了季节变化外，天气也会发生显著的短期变化。暴雨过后一两天内，土壤湿度可能发生剧烈改变，因此选择土壤研究的影像获取时间非常关键。

Another temporal aspect of importance is the comparison of *leaf-off* photography with *leaf-on* photography. Leaf-off conditions are preferable for applications in which it is important to be able to see as much detail as possible underneath trees. Such applications include activities such as topographic mapping and urban feature identification. Leaf-on conditions are preferred for vegetation mapping. Figure 1.32a illustrates leaf-on photography. Here considerable detail of the ground surface underneath the tree crowns is obscured. Figure 1.32b illustrates leaf-off photography. Here, the ground surface underneath the tree crowns is much more visible than in the leaf-on photograph. Because leaf-off photographs are typically exposed in spring or fall, there are longer shadows in the image than with leaf-on photography, which is typically exposed in summer. (Shadow length also varies with time of day.) Also, these photographs illustrate leaf-on and leaf-off conditions in an urban area

where most of the trees are deciduous and drop their leaves in fall (leaf-off conditions). The evergreen trees in the images (e.g., lower right) maintain their needles throughout the year and cast dark shadows. Hence, there would not be leaf-off conditions for such trees.

另一个重要的时间维度是对比叶盛期影像与叶枯期影像。叶枯期更适合清晰的观察树冠下方细节，例如地形测绘和城市地物识别。叶盛期则更适植被制图。图 1.32a 为叶盛期影像，树冠下方的地表细节被大量遮挡；图 1.32b 为叶枯期影像，树冠下方的地表比叶盛期影像中清晰得多。叶枯期影像通常拍摄于春季或秋季，相比夏季拍摄的叶盛期影像，阴影更长（阴影长度也会随一天中的时段变化）。除此之外，这些影像展示的是城市区域的叶盛与叶枯状态——该区域的树木多为落叶树，秋季会落叶（进入叶枯期）；而影像中的常绿树（如右下角）全年保持针叶，会投射出深色阴影，因此这类树木不存在叶枯期。



(a)



(b)

Figure 1.32 Comparison of leaf-on photography with leaf-off photography. Gladstone, OR. (a) Leaf-on photograph exposed in summer. (b) Leaf-off photograph exposed in spring. Scale 1:1,500. (Courtesy Oregon Metro)

图 1.32 俄勒冈州格拉德斯通市叶盛期影像与叶枯期影像对比图 (a) 夏季拍摄的叶盛期影像。(b) 春季拍摄的叶枯期影像。比例尺：1:1500。（图片由俄勒冈都会区政府提供）

Image Spatial Resolution and Ground Sample Distance

影像的空间分辨率与地面采样距离

Every remote sensing system has a limit on how small an object on the earth's surface can be and still be "seen" by a sensor as being separate from its surroundings. This limit, called the *spatial resolution* of a sensor, is an indication of how well a sensor can record spatial detail. In some cases, the *ground sample distance (GSD)*, or ground area represented by a single pixel in a digital image, may correspond closely to the spatial resolution of that sensor. In other cases, the ground sample distance may be larger or smaller than the sensor's spatial resolution, perhaps as a result of the A-to-D conversion process or of digital image manipulation such as resampling (see Chapter 7 and Appendix B). This distinction between spatial resolution and ground sample distance is subtle but important. For the sake of simplicity, the following discussion treats the GSD in a digital image as being equivalent to the spatial resolution of the sensor that produced the image, but note that in actual images the sampling distance may be larger or smaller than the spatial resolution.

所有的遥感系统对地表可识别物体的最小尺寸都存在限制——只有当物体能与周围环境区分开时，传感器才能“探测到”它。这种限制被称为传感器的**空间分辨率**，它直接反映了传感器记录空间细节的能力。在一些情况下，**地面采样距离（GSD）**，即数字影像中单个像素所代表的地表面积，可能与传感器的空间分辨率高度吻合。但在其他情况下，受模数转换过程、重采样等数字影像处理操作影响（详见第 7 章和附录 B），地面采样距离可能大于或小于传感器的空间分辨率。空间分辨率与地面采样距离的区别虽然很微小，但却至关重要。为了简化叙述，下文数字影像的地面采样距离与获取该影像的传感器空间分辨率表达的意义一致，但是需要注意，实际的影像中采样距离可能与空间分辨率存在差异。

Figure 1.33 illustrates, in the context of a digital image, the interplay between the spatial resolution of a sensor and the spatial variability present in a ground scene. In (a), a single pixel covers only a small area of the ground (on the order of the width of the rows of the crop shown). In (b), a coarser ground resolution is depicted and a single pixel integrates the radiance from both the crop rows and the soil between them. In (c), an even coarser resolution results in a pixel measuring the average radiance over portions of two fields. Thus, depending on the spatial resolution of the sensor and the spatial structure of the ground area being sensed, digital images comprise a range of "pure" and "mixed" pixels. In general, the larger the percentage of mixed pixels, the more limited is the ability to record and extract spatial detail in an image. This is illustrated in Figure 1.34, in which the same area has been imaged over a range of different ground resolution cell sizes.

图 1.33 以数字影像为场景，展示了传感器空间分辨率与地表场景空间变异性的相互作用：（a）中单个像素仅能代表小范围地表（大致相当于图中作物行的宽度）；（b）的地面分辨率较大，单个像素代表了作物行与行间土壤的辐射亮度；（c）的分辨率更大，单个像素代表的是两块田地部分区域的平均辐射亮度。因此，根据传感器空间分辨率及探测区域的空间结构，数字影像中会存在“纯像素”与“混合像素”。通常情况下，混合像素的占比越高，影像记录和提取空间细节的能力就越有限。图 1.34 对此进行了直观呈现，该图展示了同一区域在不同地面分辨率像元尺寸下的成像效果。

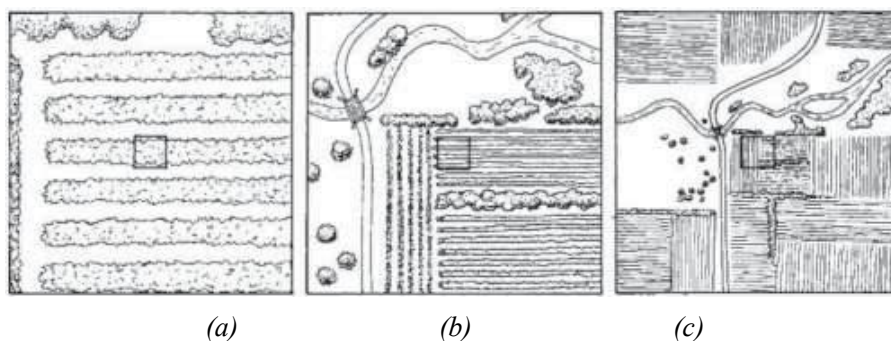


Figure 1.33 Ground resolution cell size effect: (a) small, (b) intermediate, and (c) large ground resolution cell size.

图 1.33 地面分辨率像元尺寸的影响：（a）小尺寸、（b）中尺寸、（c）大尺寸地面分辨率像元

Further discussion of the spatial resolution of remote sensing systems—including the factors that determine spatial resolution and the methods used for measuring or calculating a system's resolution—can be found in Chapter 2 (for camera systems), Chapters 4 and 5 (for airborne and spaceborne multispectral and thermal sensors), and Chapter 6 (for radar systems).

关于遥感系统空间分辨率的更多讨论——包括决定空间分辨率的因素、测量或计算系统分辨率的方法，可参阅第2章（相机系统）、第4-5章（机载和星载多光谱及热红外传感器）及第6章（雷达系统）。

Other Forms of Resolution Important in Image Interpretation

图像解译中其他重要的分辨率

It should be noted that there are other forms of resolution that are important characteristics of remote sensing images. These include the following:

需要说明的是遥感影像还有其他几种重要的分辨率，具体包括以下几种：

Spectral resolution, referring to a sensor's ability to distinguish among different ground features based on their spectral properties. Spectral resolution depends upon the number, wavelength location, and narrowness of the spectral bands in which a sensor collects image data. The bands in which any sensor collects data can range from a single broad band (for *panchromatic* images), a few broad bands (for *multispectral* images), or many very narrow bands (for *hyperspectral* images).

光谱分辨率，指传感器根据地表地物的光谱特性区分不同地物的能力。光谱分辨率取决于传感器采集影像数据时所设光谱波段的数量、波长位置及波段宽度。任何传感器的采集波段范围都不同，可分为单条宽波段（适用于**全色影像**）、多条宽波段（适用于**多光谱影像**）或多条极窄波段（适用于**高光谱影像**）。

Radiometric resolution, referring to the sensor's ability to differentiate among subtle variations in brightness. Does the sensor divide the range from the "brightest" pixel to "darkest" pixel that can be recorded in an image (the *dynamic range*) into 256, or 512, or 1024 gray level values? The finer the radiometric resolution is, the greater the quality and interpretability of an image. (See also the discussion of quantization and digital numbers in Section 1.5.)

辐射分辨率，指传感器能够区分的亮度细微变化的能力。传感器会将影像中可记录的“最亮”像素到“最暗”像素的范围（即**动态范围**）划分为256、512或

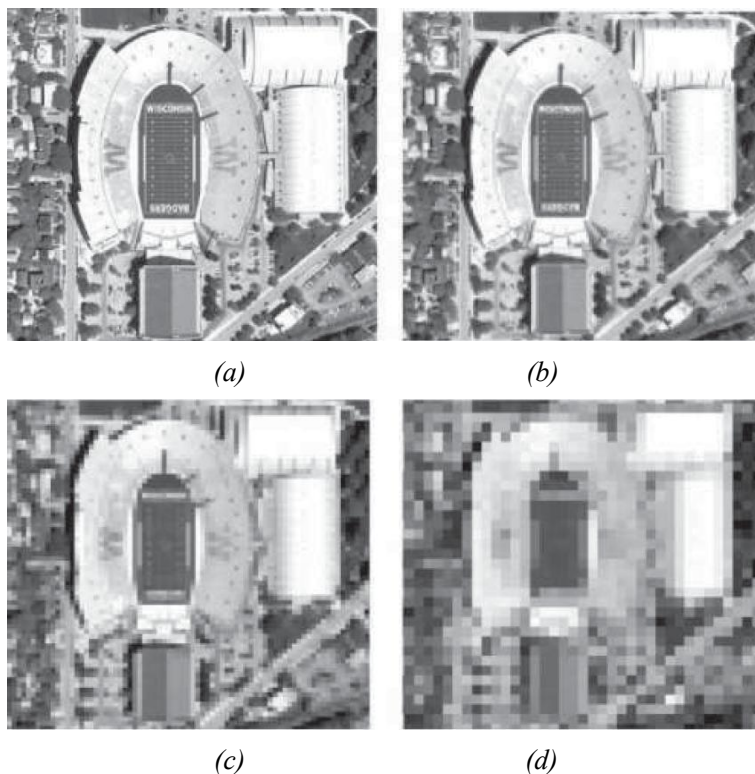
1024 个灰度级？辐射分辨率越高，影像的质量和可解译性就越强（详见 1.5 节关于量化与数字值的讨论）。

Temporal resolution, referring to the ability to detect changes over shorter or longer periods of time. Most often, this term is used in reference to a sensor that produces a time-series of multiple images. This could be a satellite system with a defined 16-day or 26-day orbital repeat cycle, or a tripod-mounted camera with a timer that collects one image every hour to serve as reference data. The importance of rapid and/or repeated coverage of an area varies dramatically with the application at hand. For example, in disaster response applications, temporal resolution might outweigh the importance of some, or all, of the other types of resolution we have summarized above.

时间分辨率，指传感器探测地表短周期或长周期变化的能力。该术语通常用于可以生成多时相影像序列的传感器，例如轨道重复周期为 16 天或 26 天的卫星系统，或是架在三脚架上、通过定时器每小时采集一次影像作为参考数据的相机。时间分辨率对某一区域进行快速、重复观测的重要性会因具体应用场景而存在显著差异。例如在灾害响应中，时间分辨率的重要性可能会超过以上所有或部分其它类型的分辨率。

In subsequent chapters, as new remote sensing systems are introduced and discussed, the reader should keep in mind these multiple resolutions that determine whether a given system would be suitable for a particular application.

在后续章节介绍和讨论新型遥感系统时，读者需牢记这些分辨率类型，它们共同决定了某一遥感系统是否适用于特定的应用场景。



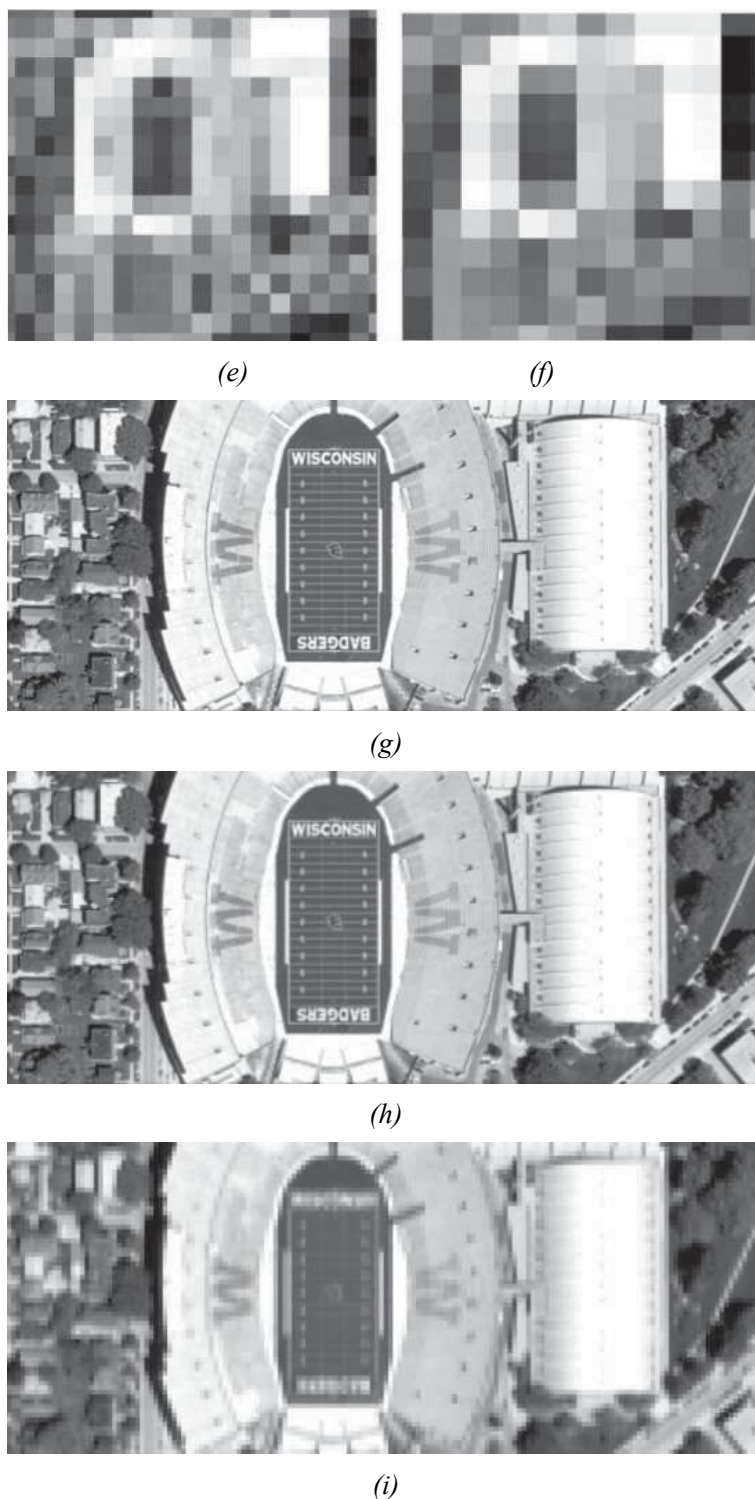


Figure 1.34 Ground resolution cell size effect on ability to extract detail from a digital image. Shown is a portion of the University of Wisconsin-Madison campus, including Camp Randall Stadium and vicinity, at a ground resolution cell size (per pixel) of: (a) 1 m, (b) 2.5 m, (c) 5 m, (d) 10 m, (e) 20 m, and (f) 30 m, and an enlarged portion of the image at (g) 0.5 m, (h) 1 m, and (i) 2.5 m. (Courtesy University of Wisconsin-Madison, Environmental Remote Sensing Center, and NASA Affiliated Research Center Program.)

图 1.34 地面分辨率像元尺寸对数字影像细节提取能力的影响，所示为威斯康星大学麦迪逊分校校园部分区域含兰德尔体育场及周边，各图对应的地面分辨率像元尺寸（每像素）如下：（a）1 米、（b）2.5 米、（c）5 米、（d）10 米、（e）20 米、（f）30 米；影像局部

放大图对应的分辨率：(g) 0.5 米、(h) 1 米、(i) 2.5 米。（图片由威斯康星大学麦迪逊分校、环境遥感中心及美国国家航空航天局（NASA）附属研究中心项目提供）

Image Scale

影像比例尺

Image scale, discussed in detail in Section 3.3, affects the level of useful information that can be extracted from aerial and space images. The scale of an image can be thought of as the relationship between a distance measured on the image and the corresponding distance on the ground. Although terminology with regard to image scale has not been standardized, we can consider that *small-scale* images have a scale of 1:50,000 or smaller, *medium-scale* images have a scale between 1:12,000 and 1:50,000, and *large-scale* airphotos have a scale of 1:12,000 or larger.

影像比例尺（详见 3.3 节）会影响从航空影像和航天影像中提取的有用信息量。影像比例尺可理解为影像上的距离与对应地表实际距离的比值。尽管影像比例尺相关术语尚未完全标准化，但通常认为：**小比例尺**影像的比例尺为 1:50,000 及更小；**中比例尺**影像的比例尺介于 1:12,000 至 1:50,000 之间；**大比例尺**航空像片的比例尺为 1:12,000 及更大。

In the case of digital data, images do not have a fixed scale per se; rather, they have a specific ground sample distance, as discussed previously (and illustrated in Figures 1.33 and 1.34), and can be reproduced at various scales. Thus, one could refer to the *display scale* of a digital image as it is displayed on a computer monitor or as printed in hardcopy.

对于数字数据而言，影像本身并没有固定比例尺，而只具有前文介绍的地面采样距离（见图 1.33 和图 1.34），而且可以按不同比例尺输出。因此，数字影像的比例尺通常指在电脑显示器上的显示比例尺，或纸质打印件的输出比例尺。

In the figure captions of this book, we have stated the hardcopy display scale of many images—including photographic, multispectral, and radar images—so that the reader can develop a feel for the degree of detail that can be extracted from images of varying scales.

本书的图表说明中，已标注了多张影像（包括摄影影像、多光谱影像和雷达影像）的纸质显示比例尺，以便读者直观感受不同比例尺影像的细节提取能力。

As generalizations, the following statements can be made about the appropriateness of various image scales for resource studies. Small-scale images are used for regional surveys, large-area resource assessment, general resource management planning, and large-area disaster assessment. Medium-scale images are used for the identification, classification, and mapping of such features as tree species, agricultural crop type, vegetation community, and soil type. Large-scale images are used for the intensive monitoring of specific items such as surveys of the damage caused by plant disease, insects, or tree blowdown. Large-scale images are also used for emergency response to such events as hazardous waste spills and planning search and rescue operations in association with tornadoes, floods, and hurricanes.

以下是不同影像比例尺在资源调查中的适用性的简单介绍：小比例尺影像适用于区域概查、大范围资源评估、资源管理总体规划及大面积灾害评估。中比例尺影像适用于树种、农作物类型、植被群落、土壤类型等地物的识别、分类与制图。大比例尺影像适用于特定对象的精细化监测，例如植物病害、虫害或树木倒

伏造成的损害调查；也可用于应急响应，如危险废物泄漏处理、龙卷风、洪水、飓风相关的搜救行动规划。

In the United States, the National High-Altitude Photography (NHAP) program, later renamed the National Aerial Photography Program (NAPP), was a federal multiagency activity coordinated by the U.S. Geological Survey (USGS). It provided nationwide photographic coverage at nominal scales ranging from 1:80,000 and 1:58,000 (for NHAP) to 1:40,000 (for NAPP). The archive of NHAP and NAPP photos has proven to be an extremely valuable ongoing source of medium-scale images supporting a wide range of applications.

在美国，“国家高空摄影计划（NHAP）”后更名为“国家航空摄影计划（NAPP）”，这是一项由美国地质调查局（USGS）协调的联邦多机构合作项目。该计划提供了覆盖全美的航空像片，NHAP 阶段的标称比例尺为 1:80,000 和 1:58,000，NAPP 阶段为 1:40,000。NHAP 和 NAPP 的影像档案已成为高质量中比例尺影像的长期获取来源，支撑着各类应用场景。

The National Agriculture Imagery Program (NAIP) acquires peak growing season leaf-on imagery in the continental United States and delivers this imagery to U.S. Department of Agriculture (USDA) County Service Centers in order to assist with crop compliance and a multitude of other farm programs. NAIP imagery is typically acquired with GSDs of one to two meters. The one-meter GSD imagery is intended to provide updated digital orthophotography. The two-meter GSD imagery is intended to support USDA programs that require current imagery acquired during the agricultural growing season but do not require high horizontal accuracy. NAIP photographs are also useful in many non-USDA applications, including real estate, recreation, and land use planning.

“国家农业影像计划（NAIP）”在美国本土的植被生长旺季采集叶盛期影像，并将其提供给美国农业部（USDA）的服务中心，用于辅助作物合规性核查及其他多项农业相关项目。NAIP 影像的地面采样距离（GSD）通常为 1 至 2 米：1 米 GSD 影像用于更新数字正射影像；2 米 GSD 影像用于支持 USDA 各类项目开展，这类项目需获取农业生长季的最新影像，但对水平精度要求不高。此外，NAIP 影像还广泛应用于非 USDA 领域，包括房地产、休闲娱乐及土地利用规划等。

Approaching the Image Interpretation Process

影像解译流程的实施方法

There is no single "right" way to approach the image interpretation process. The specific image products and interpretation equipment available will, in part, influence how a particular interpretation task is undertaken. Beyond these factors, the specific goals of the task will determine the image interpretation process employed. Many applications simply require the image analyst to identify and count various discrete objects occurring in a study area. For example, counts may be made of such items as motor vehicles, residential dwellings, recreational watercraft, or animals. Other applications of the interpretation process often involve the identification of anomalous conditions. For example, the image analyst might survey large areas looking for such features as failing septic systems, sources of water pollution entering a stream, areas of a forest stressed by an insect or disease problem, or evidence of sites having potential archaeological significance.

影像解译不存在唯一正确的方法。可供使用的特定影像产品和解译设备，会在一定程度上影响具体解译任务的执行方式。除此之外，任务的具体目标会决定

所使用的影像解译流程。许多应用场景仅要求影像分析师识别并统计研究区域内的各类离散目标。例如，统计机动车、住宅建筑、休闲船只或动物等对象的数量。另一些解译应用则常涉及异常情况的识别，比如分析师可能会巡查大范围区域，寻找化粪池故障、溪流污染源、受病虫害影响的林区，或具有潜在考古价值的遗址痕迹。

Many applications of image interpretation involve the delineation of discrete areal units throughout images. For example, the mapping of land use, soil types, or forest types requires the interpreter to outline the boundaries between areas of one type versus another. Such tasks can be problematic when the boundary is not a discrete edge, but rather a "fuzzy edge" or gradation from one type of area to another, as is common with natural phenomena such as soils and natural vegetation.

影像解译的更多应用还包括在影像中勾勒离散的面状单元边界。例如，土地利用、土壤类型或森林类型制图时，解译者需要划分不同类型区域之间的界限。但当边界并非清晰的离散边缘，而是呈现模糊边缘，或从一种类型向另一种类型渐变时，这类任务会面临挑战，土壤、自然植被等自然现象通常都是如此。

Two extremely important issues must be addressed before an interpreter undertakes the task of delineating separate areal units on remotely sensed images. The first is the definition of the *classification system* or criteria to be used to separate the various categories of features occurring in the images. For example, in mapping land use, the interpreter must fix firmly in mind what specific characteristics determine if an area is "residential," "commercial," or "industrial." Similarly, the forest type mapping process must involve clear definition of what constitutes an area to be delineated in a particular species, height, or crown density class.

在解译者在遥感影像上勾勒离散面状单元之前，必须解决两个非常重要的问题。首先是确定**分类体系**或判据，用于区分影像中出现的各类地物。例如，进行土地利用制图时，解译者必须明确界定哪些具体特征能将某一区域判定为“住宅用地”、“商业用地”或“工业用地”；同样，森林类型制图也需要清晰定义，什么样的区域应该被划分为特定树种、树高或树冠的密度等级。

The second important issue in delineation of discrete areal units on images is the selection of the *minimum mapping unit* (MMU) to be employed in the process. This refers to the smallest size areal entity to be mapped as a discrete area. Selection of the MMU will determine the extent of detail conveyed by an interpretation. This is illustrated in Figure 1.35. In (a), a small MMU results in a much more detailed interpretation than does the use of a large MMU, as illustrated in (b).

第二个重要的问题是选择**最小制图单元（MMU）**。它指的是能够作为离散区域进行制图的最小面积实体。最小制图单元的选择将决定解译结果所呈现的详细程度，如图 1.35 所示：（a）中使用小尺度最小制图单元，解译结果比（b）中使用大尺度最小制图单元的结果详细得多。

Once the classification system and MMU have been determined, the interpreter can begin the process of delineating boundaries between feature types. Experience suggests that it is advisable to delineate the most highly contrasting feature types first and to work from the general to the specific. For example, in a land use mapping effort, it would be better to separate "urban" from "water" and "agriculture" before separating more detailed categories of each of these feature types based on subtle differences.

确定分类体系和最小制图单元后，解译者就可以开始划分不同地物类型之间的边界。根据实践经验，建议先勾勒对比度最强的地物类型，遵循从整

体到局部的原则。例如，进行土地利用制图时，应先将“城镇用地”与“水域”、“农业用地”区分开，再根据细微差异划分各类地物内部更细微的类别。

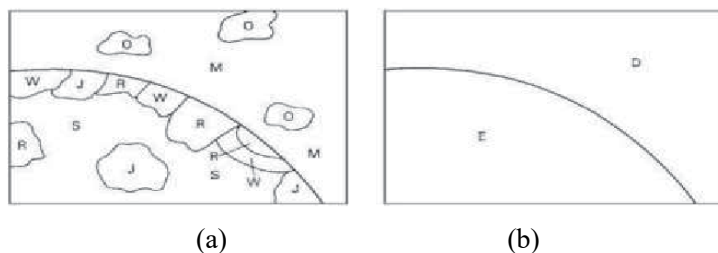


Figure 1.35 Influence of minimum mapping unit size on interpretation detail. (a) Forest types mapped using a small MMU: O, oak; M, maple; W, white pine; J, jack pine; R, red pine; S, spruce. (b) Forest types mapped using a large MMU: D, deciduous; E, evergreen.

图 1.35 最小制图单元大小对解译详细程度的影响。(a) 使用小尺度最小制图单元绘制的森林类型：O-栎树；M-枫树；W-白皮松；J-黑松；R-赤松；S-云杉。(b) 使用大尺度最小制图单元绘制的森林类型：D-落叶林；E-常绿树。

In certain applications, the interpreter might choose to delineate *photomorphic regions* as part of the delineation process. These are regions of reasonably uniform tone, texture, and other image characteristics. When initially delineated, the feature type identity of these regions may not be known. Field observations or other ground truth can then be used to verify the identity of each region. Regrettably, there is not always a one-to-one correspondence between the appearance of a photomorphic region and a mapping category of interest. However, the delineation of such regions often serves as a stratification tool in the interpretation process and can be valuable in applications such as vegetation mapping (where photomorphic regions often correspond directly to vegetation classes of interest).

在某些应用场景中，解译者可能会在勾勒过程中划分**影像形态区域**。这类区域具有相对均匀的色调、纹理及其他影像特征。初步划分时，这些区域的地物类型可能尚不明确，后续可通过实地观测或其他地面数据验证每个区域的属性。遗憾的是，影像形态区域的表现与目标制图类别之间并不是总能形成一一对应关系。但划分这类区域通常可作为解译过程中的分层过程，在植被制图等应用中具有重要价值（这类场景中影像形态区域往往与目标植被类别直接对应）。

Basic Equipment for Visual Interpretation

目视解译的基础设备

Visual image interpretation equipment generally serves one of several fundamental purposes: viewing images, making measurements on images, performing image interpretation tasks, and transferring interpreted information to base maps or digital databases. Basic equipment for viewing images and transferring interpreted information is described here. Equipment involved in performing measuring and mapping tasks will be described in Chapter 3.

目视影像解译设备通常有以下几个核心功能：影像查看、影像量测、解译任务执行，以及将解译信息转绘至底图或数字数据库。本文将介绍用于影像查看和信息转绘的基础设备，量测与制图相关设备将在第3章详细说明。

The airphoto interpretation process typically involves the utilization of stereoscopic viewing to provide a three-dimensional view of the terrain. Some space images are also analyzed stereoscopically. The stereo effect is possible because we have binocular vision. That is, since we have two eyes that are slightly separated, we continually view the world

from two slightly different perspectives. Whenever objects lie at different distances in a scene, each eye sees a slightly different view of the objects. The differences between the two views are synthesized by the mind to provide depth perception. Thus, the two views provided by our separated eyes enable us to see in three dimensions.

航空像片的解译过程通常需借助立体观测获取地形的三维视图，部分航天影像也会采用立体分析的方式。立体效果的实现是基于人的双眼视觉原理：由于人的双眼存在轻微间距，我们始终从两个有差异的视角观察世界。当场景中物体处于不同距离时，每只眼睛看到的物体影像会存在细微差别，大脑通过整合这两组差异影像形成深度感知，进而形成了三维视觉。

Vertical aerial photographs are often taken along flight lines such that successive images overlap by at least 50% (see Figure 3.2). This overlap also provides two views taken from separated positions. By viewing the left image of a pair with the left eye and the right image with the right eye, we obtain a three-dimensional view of the terrain surface. The process of stereoscopic viewing can be done using a traditional *stereoscope*, or using various methods for stereoscopic viewing on computer monitors. This book contains many *stereopairs*, or *stereograms*, which can be viewed in three dimensions using a lens stereoscope such as shown in Figure 1.36. An average separation of about 58 mm between common points has been used in the stereograms in this book. The exact spacing varies somewhat because of the different elevations of the points.

垂直航空像片通常沿航线拍摄，相邻影像的重叠度至少为 50%（见图 3.2）。这种重叠设计使同一区域获得两个不同拍摄位置的影像，通过左眼观察像对中的左片、右眼观察右片，即可呈现地形表面的三维效果。立体观测可通过传统**立体镜**完成，也可借助计算机显示器等多种立体观测技术实现。本书包含大量**立体像对（或称立体图）**，使用图 1.36 所示的透镜式立体镜即可进行三维观测。书中立体图的同名点间距平均约为 58 毫米，因点位海拔不同，实际间距会略有差异。

Typically, images viewed in stereo manifest *vertical exaggeration*. This is caused by an apparent difference between the vertical scale and the horizontal scale of the stereomodel. Because the vertical scale appears to be larger than the horizontal scale, objects in the stereomodel appear to be too tall. A related consideration is that slopes in the stereomodel will appear to be steeper than they actually are. (The geometric terms and concepts used in this discussion of vertical exaggeration are explained in more detail in Chapter 3.)

通常，立体观测状态下的影像会呈现**垂直夸大现象**，这是由立体模型的垂直比例尺与水平比例尺存在差异导致的。由于垂直比例尺通常大于水平比例尺，立体模型中的物体显得比实际更高，地形坡度也会比实际的更陡峭（关于垂直夸大讨论中涉及的几何术语与概念，将在第 3 章展开详细解释）。



Figure 1.36 Simple lens stereoscope. (Author-prepared figure.)

图 1.36 简易透镜式立体镜（作者自制）。

Many factors contribute to vertical exaggeration, but the primary cause is the lack of equivalence between the original, in-flight, *photographic base–height ratio*, B/H' (Figure 3.24a), and the corresponding, in-office, *stereoviewing base–height ratio*, b_e/h_e (Figure 1.37). The perceived vertical exaggeration in the stereomodel is approximately the ratio of these two ratios. The photographic base–height ratio is the ratio of the *air base* distance between the two exposure stations to the flying height above the average terrain elevation. The stereoviewing base–height ratio is the ratio between the viewer's *eye base* (b_e) to the distance from the eyes at which the stereomodel is perceived (h_e). The perceived vertical exaggeration, VE , is approximately the ratio of the photographic base–height ratio to the stereoviewing base–height ratio,

很多的原因都会导致垂直夸大，但主要原因是飞行中原始的**摄影基线-航高比** B/H' （图 3.24a）与室内立体观测时对应的**观测基线-视高比** b_e/h_e （图 1.37）不相等。立体模型观测中感知到的垂直夸大程度约等于这两个比值之比。摄影基线-航高比指两个焦点之间的**空中基线**距离与平均地形高程以上飞行高度的比值。立体观测基线-视高比指观测者的**眼基线**（ b_e ）与立体模型感知距离（ h_e ）的比值。感知垂直夸大系数 VE 约等于摄影基线-航高比与立体观测基线-视高比的比值，即：

$$VE \cong \frac{B/H'}{b_e/h_e} \quad (1.10)$$

where $b_e/h_e = 0.15$ for most observers.

其中，大多数观测者的 $b_e/h_e = 0.15$ 。

In short, vertical exaggeration varies directly with the photographic base–height ratio.

简而言之，垂直夸大程度与摄影基线-航高比比呈正比。

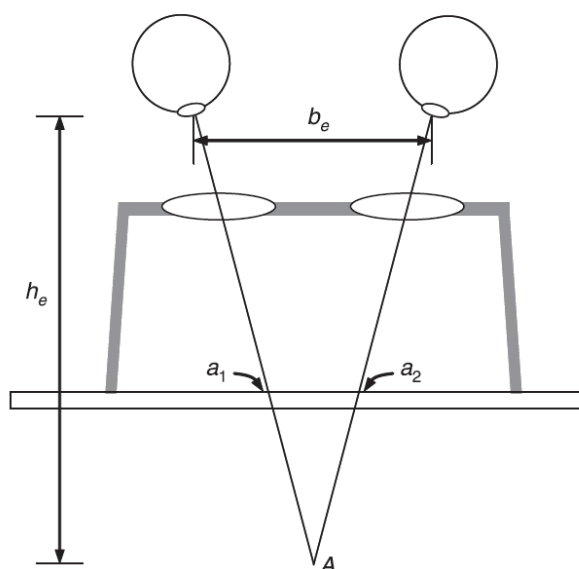


Figure 1.37 Major cause of vertical exaggeration perceived when viewing a stereomodel. There is a lack of equivalence between the photographic base–height ratio when the photographs are taken and the stereoviewing base–height ratio when the photographs are viewed with a stereoscope.

图 1.37 观测立体模型时感知到垂直夸大的主要原因：摄影拍摄时的摄影基线-航高比，与通过立体镜观测照片时的立体观测基线-视高比不相等。

While vertical exaggeration is often misleading to novice image interpreters, it is often very useful in the interpretation process. This is due to the fact that subtle variation in the heights of image features is more readily discriminated when exaggerated. As we note in Chapter 3, large base-height ratios also improve the quality of many photogrammetric measurements made from vertical aerial photographs.

虽然垂直夸大常常让影像解译新手产生误解，但它在解译过程中往往十分实用。这是因为地物高度的细微变化经过夸大后，更容易被区分和识别。正如第 3 章所写的那样，较大的基线-航高比能提升基于垂直航空像片的多项摄影测量精度。

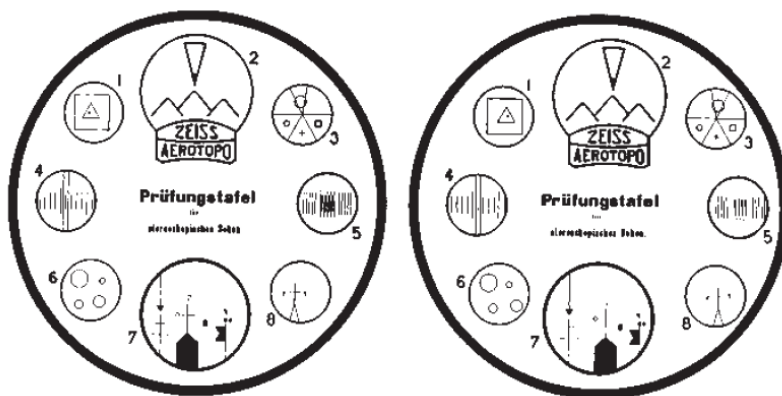


Figure 1.38 Stereoscopic vision test. (Courtesy Carl Zeiss, Inc.)

图 1.38 立体视觉测试图（卡尔·蔡司公司提供）。

Figure 1.38 can be used to test stereoscopic vision. When this diagram is viewed through a stereoscope, the rings and other objects should appear to be at varying distances from the observer. Your stereovision ability can be evaluated by filling in Table 1.3 (answers are in the second part of the table). People whose eyesight is very weak in one eye may not have the ability to see in stereo. This will preclude three-dimensional viewing of the stereograms in this book. However, many people with essentially monocular vision have become proficient photo interpreters. In fact, many forms of interpretation involve monocular viewing with such basic equipment as handheld magnifying glasses or tube magnifiers (2x to 10x lenses mounted in a transparent stand).

图 1.38 可以用来测试观察者的立体视觉能力。通过立体镜观测该图时，圆环及其他物体呈现出与观测者距离不同的立体效果。你可通过填写表 1.3 评估自身立体视觉能力（答案见表格后半部分）。单眼视力非常差的人可能无法形成立体视觉，这会影响本书中立体图的三维观测效果。但事实上，许多单眼视觉功能基本正常的人也能成为熟练的影像解译者，不少解译工作只需要借助手持放大镜或管式放大镜（2 倍至 10 倍透镜安装在透明支架上）等基础设备，通过单眼观测即可完成。

TABLE 1.3 Stereovision Test for Use with Figure 1.38

表 1.3 配合图 1.38 使用的立体视觉测试表

PART I (第一部分)

Within the rings marked 1 through 8 are designs that appear to be at different elevations. Using “1” to designate the highest elevation, write down the depth order of the designs. It is possible that two or more designs may be at the same elevation. In this case, use the same number for all designs at the same elevation.

在标有数字 1 至 8 的圆环内，各图案呈现出不同的深度效果。以数字“1”表示最高深度，写下这些图案的深度排序。若两个或多个图案处于同一深度，可对所有同深度图案标注相同数字。

Ring 1 (圆环 1)

Square (正方形)

Marginal ring (边缘环)

Triangle (三角形)

Point (点)

Ring 6 (圆环 6)

Lower left circle (左下圆)

Lower right circle (右下圆)

Upper right circle (右上圆)

Upper left circle (左上圆)

Marginal ring (边缘环)

Ring 7 (圆环 7)

Black flag with ball (带球的黑旗)

Marginal ring (边缘环)

Black circle (黑圆)

Arrow (箭头)

Tower with cross (带十字的塔形)

Double cross (双十字)

Black triangle (黑三角形)

Black rectangle (黑矩形)

Ring 3 (圆环 3)

Square (正方形)

Marginal ring (边缘环)

Cross (十字)

Lower left circle (左下圆)

Upper left circle (左上圆)

PART II (第二部分)

Indicate the relative elevations of the rings 1 through 8.

请标注圆环 1 至 8 的相对深度。

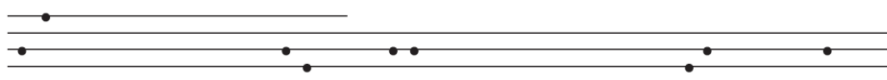
() () () () () () ()
(最高) Highest Lowest (最低)

PART III (第三部分)

Draw profiles to indicate the relative elevations of the letters in the words “prüfungstafel” and “stereoskopisches sehen.”

绘制剖面图，标注“prüfungstafel”和“stereoskopisches sehen”这两个词中各字母的相对深度。

P R U F U N G S T A F E L S T E R E O S K O P I S C H E S S E H E N



(Answers to Stereovision Test on next page)

(立体视觉测试答案见下一页)

TABLE 1.3 (Continued)

表 1.3 (续表)

PART I (第一部分)

Ring 1 (圆环 1)

Square (正方形)	(2)
Marginal ring (边缘环)	(1)
Triangle (三角形)	(3)
Point (点)	(4)

Ring 6 (圆环 6)

Lower left circle (左下圆)	(4)
Lower right circle (右下圆)	(5)
Upper right circle (右上圆)	(1)
Upper left circle (左上圆)	(3)
Marginal ring (边缘环)	(2)

Ring 7 (圆环 7)

Black flag with ball (带球的黑旗)	(5)
Marginal ring (边缘环)	(1)
Black circle (黑圆)	(4)
Arrow (箭头)	(2)
Tower with cross (带十字的塔形)	(7)
Double cross (双十字)	(2)
Black triangle (黑三角形)	(3)
Black rectangle (黑矩形)	(6)

Ring 3 (圆环 3)

Square (正方形)	(4)
Marginal ring (边缘环)	(2)
Cross (十字)	(3)
Lower left circle (左下圆)	(1)
Upper left circle (左上圆)	(5)

PART II (第二部分)

(7) (6) (5) (1) (4) (2) (3) (8)
(最高) Highest Lowest (最低)

PART III (第三部分)



Some people will be able to view the stereograms in this book without a stereoscope. This can be accomplished by holding the book about 25 cm from your eyes and allowing the view of each eye to drift into a straight-ahead viewing position (as when looking at objects at an infinite distance) while still maintaining focus on the stereogram. When the two images have fused into one, the stereogram will be seen in three dimensions. Most persons will find stereoviewing without proper stereoscopes to be a tiring procedure, producing eyestrain. It is, however, a useful technique to employ when stereoscopes are not available.

部分人无需立体镜也能观测本书中的立体图。将书本置于距眼睛约 25 厘米处，双眼视线自然放空（如同注视无限远处物体），同时保持对立体图的对焦。当两张影像融合为一张时，就能看到三维立体效果。但多数人会发现，没有专业立体镜辅助的立体观测过程容易疲劳而引发眼部不适。不过在无立体镜的情况下，这仍是一种实用的应急观测方法。

Several types of analog stereoscopes are available, utilizing lenses or a combination of lenses, mirrors, and prisms. *Lens stereoscopes*, such as the one shown in Figure 1.36, are portable and comparatively inexpensive. Most are small instruments with folding legs. The lens spacing can usually be adjusted from about 45 to 75 mm to accommodate individual eye spacings. Lens magnification is typically 2 power but may be adjustable. The principal disadvantage of small lens stereoscopes is that the images must be quite close together to be positioned properly underneath the lenses. Because of this, the interpreter cannot view the entire stereoscopic area contained by the overlapping aerial photographs without raising the edge of one of the photographs. More advanced devices for viewing hardcopy stereoscopic image pairs include *mirror stereoscopes* (larger stereoscopes that use a combination of prisms and mirrors to separate the lines of sight from each of the viewer's eyes) and *zoom stereoscopes*, expensive precision instruments used for viewing stereopairs under variable magnification.

模拟立体镜的类型非常多样，核心组件包括透镜，或透镜、反射镜与棱镜的组合物。图 1.36 所示的**透镜式立体镜**便携且价格相对低廉，大部分为带折叠支架的小型设备。透镜间距通常可以在 45 至 75 毫米之间调节，以适配不同人的瞳距，放大倍数一般为 2 倍，部分可调节。这类小型透镜式立体镜的主要缺点是，两张影像需紧密贴合才能准确对准透镜下方，因此观测者若不掀开其中一张像片的边缘，无法完整查看航空像片重叠区域的全部立体范围。用于观测纸质立体像对更先进的设备有**反射式立体镜**（体积较大，通过棱镜与反射镜组合来分离双眼视线）和**变焦立体镜**，高精度、昂贵的设备可以在可变放大倍数下观测立体像对。

With the proliferation of digital imagery and software for viewing and analyzing digital images, analog stereoscopes have been replaced in the laboratory (if not in the field) by various computer hardware configurations for stereoviewing. These devices are discussed in Section 3.10.

随着数字影像的普及以及数字影像查看与分析软件的发展，模拟立体镜已经在实验室中（即便在野外尚未完全替代）被各类计算机立体观测硬件所取代。这类设备将在 3.10 节进行详细探讨。

Relationship between Visual Image Interpretation and Computer Image Processing

目视影像解译与计算机影像处理的关系

In recent years, there has been increasing emphasis on the development of quantitative, computer-based processing methods for analyzing remotely sensed data. As will be discussed in Chapter 7, those methods have become increasingly sophisticated and powerful. Despite these advances, computers are still somewhat limited in their ability to evaluate many of the visual “clues” that are readily apparent to the human interpreter, particularly those referred to as image *texture*. Therefore, visual and numerical techniques should be seen as complementary in nature, and consideration must be given to which approach (or combination of approaches) best fits a particular application.

近年来，基于计算机的定量遥感数据处理方法开发日益受到重视。正如第 7 章将要讨论的，这类方法已经变得越发精密且功能强大。尽管取得了进展，计算机在识别许多人类解译者极易察觉的视觉“线索”方面仍存在一定局限，尤其是

被影像纹理的特征。因此，目视解译技术与数值处理技术本质上应被视为互补关系，需根据具体应用场景选择最适配的单一方法或组合方法。

The discussion of visual image interpretation in this section has of necessity been brief. As mentioned at the start of this section, the skill of image interpretation is best learned interactively, through the experience of interpreting many images. In the ensuing chapters of this book, we provide many examples of remotely sensed images, from aerial photographs to synthetic aperture radar images. We hope that the reader will apply the principles and concepts discussed in this section to help interpret the features and phenomena illustrated in those images.

本节对目视影像解译的讨论难免简略。如本节开篇所述，影像解译技能掌握的最佳途径是通过解读大量影像的实践进行交互式学习。在本书后续章节中，我们将提供从航空像片到合成孔径雷达影像的各类遥感影像示例，希望读者能运用本节所讨论的原理与概念，辅助解读这些影像中呈现的地物特征与现象。

CHAPTER

2 ELEMENTS OF PHOTOGRAPHIC SYSTEMS

第二章 摄影系统的构成要素

2.1 INTRODUCTION

2.1 引言

One of the most common, versatile, and economical forms of remote sensing is aerial photography. Historically, most aerial photography has been film-based. In recent years, however, digital photography has become the most dominant form of newly collected photographic imagery. In this chapter, unless otherwise specified, when we use the term "aerial photography," we are referring to both film and digital aerial photography. The basic advantages that aerial photography affords over on-the-ground observation include:

航空摄影是最常见、用途最广泛且经济高效的遥感手段之一。历史上，大多数航空摄影以胶片为载体。但近年来，数字摄影已成为新采集摄影影像的主流形式。本章中，除非另有说明，“航空摄影”一词均指代胶片航空摄影与数字航空摄影。相较于地面观测，航空摄影的核心优势包括：

1. Improved vantage point. Aerial photography gives a bird's-eye view of large areas, enabling us to see earth surface features in their spatial context. In short, aerial photography permits us to look at the "big picture" in which objects of interest reside. It is often difficult, if not impossible, to obtain this view of the environment through on-the-ground observation. With aerial photography, we also see the "whole picture" in that *all* observable earth surface features are recorded simultaneously.

1. 观测视角的提升。航空摄影能提供大范围区域的鸟瞰视图，让我们在空间语境中观察地表地物。简而言之，它能呈现目标物体所处的“全景格局”。这种环境视角通过地面观测往往难以实现，甚至完全无法获得。同时，航空摄影可同步记录所有可观测的地表地物，实现“全貌呈现”。